Side Activity 2: Basic Formatting & GUI Tools

CSCI3170 Introduction to Database Systems

Learning Objectives

In this Side Activity, we will focus on **Data Presentation** and **Tool Proficiency**:

- Transform raw, messy SQL output into readable reports using CLI dot-commands.
- Handle data truncation issues (when text gets cut off).
- **DB Browser: A GUI interface to work on your databases**

Part 1: The Problem of "Ugly" Output

In Side Activity 1, you learned how to connect to the database. Let's quickly reconnect.

Quick Setup: Open your terminal, navigate to your folder, and run:

```
sqlite3 chinook.db
```

The Scenario

Imagine your manager asks for a list of employees. You run this standard query:

```
SELECT FirstName, LastName, Title, Email FROM employees LIMIT 3;
```

Observation: The output is likely separated by pipes (|) and difficult to read. It looks like raw data, not a report.

Part 2: Formatting Reports in CLI

To make data readable for humans, we use SQLite "dot-commands".

Step 1: Enabling Columns and Headers

Transform the output into a table format. Run these two commands:

```
sqlite> .mode column
sqlite> .headers on
```

Now, press the **Up Arrow** key to find your previous `SELECT` query and run it again. *Result: It should look like a table now, but you might notice a new problem...*

Step 2: Fixing Truncated Data (.width)

Key Concept: Why is data cut off?

By default, SQLite guesses how wide columns should be (usually 10 characters). If a Title is "Sales Support Agent", it might show as "Sales Supp...".

We need to manually tell SQLite how much space to give each column.

1. Look at your previous output. Count how many columns you selected (FirstName, LastName, Title, Email = 4 columns).
2. Set specific widths for these 4 columns (e.g., 10, 10, 20, 25 characters):

```
sqlite> .width 10 10 20 25
```

3. Run the query again:

```
SELECT FirstName, LastName, Title, Email FROM employees LIMIT 3;
```

Success! You should now see the full text for the Job Titles and Emails.

Step 3: Exporting to CSV (Optional but Useful)

If your manager wants to open the data in Excel, "Column mode" is bad. You need "CSV mode".

```
sqlite> .mode csv
sqlite> SELECT FirstName, LastName, Email FROM employees LIMIT 3;
```

Note: Remember to switch back to `.mode column` for the rest of the Side Activity!

Part 3: Using DB Browser (GUI)

Sometimes writing SQL is too slow for quick checks. **DB Browser for SQLite** allows us to "browse" data visually.

Step 0: Installation

Before we can use the GUI, you need to download and install the software:

1. Visit the official download page: `https://sqlitebrowser.org/dl/`
2. Select the version appropriate for your operating system (Windows, macOS, or Linux).
3. Download and run the installer.

Step 1: Open the Database

1. Launch **DB Browser for SQLite**. 2. Click **Open Database** and select your `chinook.db` file.

Step 2: Using Filters (The Power Feature)

Go to the **Browse Data** tab.

1. **Select the Table:** Look at the drop-down menu labeled "Table" (top-left corner). Change it from `albums` to `tracks`.
2. **Locate Filters:** You will now see many columns. Under each header, there is an empty box. These are the **Filter Boxes**.

Try these specific tasks in the GUI:

1. **Exact Match:** Scroll to the right to find the `Composer` column. In the filter box, type:

   ```
   AC/DC
   ```

   This shows only songs composed by AC/DC. *(Remember to clear this box after checking the result!)*

2. **Partial Match:** In the `Name` column filter box, type:

   ```
   %Love%
   ```

   The `%` symbol means "anything". This finds "Love Gun", "Whole Lotta Love", etc. *(Clear this box before the next step.)*

3. **Numeric Range:** In the `Milliseconds` filter box, type:

   ```
   > 300000
   ```

   This filters for songs longer than 5 minutes (300,000ms).

Part 4: Self-Exercises (not graded)

Scenario: The End-of-Month Report

You are the Database Administrator for the Chinook Music Store. You have received two urgent requests. You must choose the right tool (CLI or GUI) to solve them.

Challenge 1: The "Clean" Printed List (Use CLI)

The Store Manager wants a printed list of all **Customers** who live in the **USA**. The list must be formatted neatly so it can be pasted into a Word document.

Your Task: 1. Using the `sqlite3` terminal, configure the output to be in `column` mode with `headers` on. 2. Adjust the `.width` so that the **FirstName**, **LastName**, and **Email** fit perfectly (e.g., try widths of 12, 12, and 30). 3. Write and execute the query to find USA customers. 4. *Check:* Ensure no emails are cut off (truncated).

Challenge 2: The Quick Audit (Use GUI)

The Accounting Department suspects there are some very expensive invoices that need manual review. They don't need a report; they just want to see the data quickly.

Your Task: 1. Open DB Browser and go to the `invoices` table. 2. Use the Filter boxes to find invoices where:

- The `BillingCountry` is **Germany**.
- AND the `Total` is greater than **10.00**.

3. *Question:* How many invoices match these criteria? (Find the value at the bottom status bar of the window).

End of Side Activity 2.